

Part III

STATISTICS IN RESEARCH

Much of a researcher's time is spent accumulating and analysing data: data about temperatures, occurrences, concentrations, durations, etc. The field of statistics deals with the analysis of data.

Almost always one has imperfect data about a small subset (the sample) of the intended population. The data is *imperfect* because of experimental, recording and other errors. One has data only about a *subset* of the population because it is impossible to examine the entire population. A botanist who proposes a law of genetics can only test this law on a tiny fraction of the plants on the planet (to say nothing about those plants not yet growing); sooner or later one has to stop and conclude that the patterns found in the sample are also found in the whole.

We discussed earlier how to find a suitable sample. In the chapters on statistics we examine how to analyse the sample data and draw conclusions. This is known as **statistical inference**.

These chapters proceed as follows. We start with data and how to organize it. We then look at probability and probability distributions especially the normal distribution. This brings us to the main purpose of this part: explaining estimation and statistical tests. These are used to answer questions such as: how confidently can one extrapolate from the sample data, and does the sample data agree with a preconceived theory.

In this book we examine only the situations a researcher is most likely to encounter. If you need more information then you should consider a statistics book (e.g. Net82, Hog93, Fre90, Moo85, Mul87, Wad90) or a friendly statistician. A computer statistics package can also be useful, both for performing the necessary calculations and for representing the data in various ways. An example of a full statistics package is the SAS program; more modest packages include SPSS and Minitab. A spreadsheet program (such as QuattroPro, Excel or Lotus 1-2-3) can also be useful.

8 Basic Statistics: Data Organization

In this chapter we introduce common statistical measures and discuss the representation and organisation of data. The chapter concludes with a discussion of regression, which is a technique for fitting curves to data.

8.1 STATISTICAL MEASURES

Data can be qualitative or quantitative. **Quantitative data** has numerical values, e.g. in the range 0 to 100. **Qualitative data** has values that fall into categories, e.g. animal, vegetable or mineral. For example, whether it rained or not yesterday is a qualitative question (the categories are rained and not rained); how much rain was measured is a quantitative question. Quantitative data can be **discrete** or **continuous**: discrete if it takes on only whole values and continuous if it takes on any real value in some interval. The amount of rain yesterday is continuous data; on how many days it rained last year is discrete data.

A statistical measure, or simply a **statistic**, is a summary of the data. This data can be for the entire population or for the chosen sample. We now discuss some population statistics.

8.1.1 Measures of central tendency

Most statistics of quantitative data are either measures of central tendency (where the data is centred) or measures of dispersion (how spread out the data is). In the former category are the mean, the median and the mode.

The population **mean** is the average and is denoted by the Greek letter μ . If the population data values are $x_1, x_2, \dots, x_n$, then:

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i$$

The symbol Σ (pronounced ‘sigma’) means ‘the sum of’, so the above formula says that

$$\mu = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

The **median** is the middle value were the data to be sorted from smallest to biggest. If there is an even number of values, then the median is usually taken to be the average of the middle two values. The **mode** is defined as the most common value(s) in the data set.

Example: For the data 5, 3, 6, 7, 5, 5, 9, 1, 17, 9,
the mean is 6.7,
the median is 5.5, and
and the mode is 5.

Aside: In this book we use the decimal point in real numbers instead of the comma (e.g. 5.42 rather than 5,42).

While the mean is more commonly used, the median can be a better summary of the data if there are extreme values. For example, in statistics on the price of houses in an area, the median is often given, since the mean is very much influenced by a large mansion being sold. Say selling prices in thousands of rands were 100, 100, 110, 120, 130, 140, and 980. The mean here is 240 and the median is 120. The median is a better summary as most people paid around 120 thousand rand for their house.

For qualitative variables (e.g. colour) the mode is the only possible measure. For ranked data (e.g. 1st, 2nd, 3rd) both the median and the mode are possible, but the median is preferred as it considers the rankings rather than just frequencies.

8.1.2 Measures of dispersion

In summarising quantitative data, one is also interested in measures of dispersion, that is, how spread out the data is. Examples of these are the range, percentiles, the variance and the standard deviation.

One simple statistic is the **range**: the gap between the largest and smallest values. This is seldom useful, as there can be eccentric data. For example, the annual incomes of South Africans range from zero to many million rands—however, the spread of incomes has very few people at the top extreme and very many at the bottom, and most incomes lie in a much narrower range.

One is more likely to use **percentiles**. For example, the 90th percentile is the value below which lies 90% of the data. In this terminology, the median is the 50th percentile.

Another measure of dispersion is the **variance**; the variance of the population is denoted by σ^2 . This is defined as the average squared deviation: if the data values are $x_1, x_2, \dots, x_n$, then

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

The (positive) square-root of the data is called the **standard deviation** of the data; the population standard deviation is denoted by σ . The standard deviation is the most used measure of dispersion.

Example: For the data 5, 3, 6, 7, 5, 5, 9, 1, 17, 9 (with $\mu = 6.7$),
the range is 16,
the variance is 17.21, and
the standard deviation is 4.15.

One property of the mean and the standard deviation is that if the data comes from a ‘reasonable’ population (to be defined later), then about 68% of the data lies between $\mu - \sigma$ and $\mu + \sigma$ (that is, within one standard deviation of the mean), while 95% of the data lies within two standard deviations of the mean.

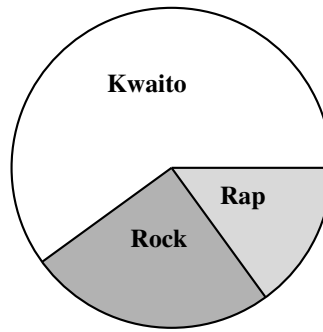
8.2 GRAPHICAL REPRESENTATION OF DATA

Sometimes a picture explains the situation much more clearly than a jumble of numbers. However, do not include a picture just to impress people; pictures should be relevant and useful. There are several common types of pictures. Spreadsheets and many other computer packages contain inbuilt graphing functions.

8.2.1 Depiction of qualitative data

A common device for displaying qualitative data is a pie chart. For example, if 60% of a country’s music exports is kwaito, 25% rock and 15% rap, then this could be depicted by the pie chart in Figure 1.

Figure 1. An example of a pie-chart.



8.2.2 Depiction of frequency data

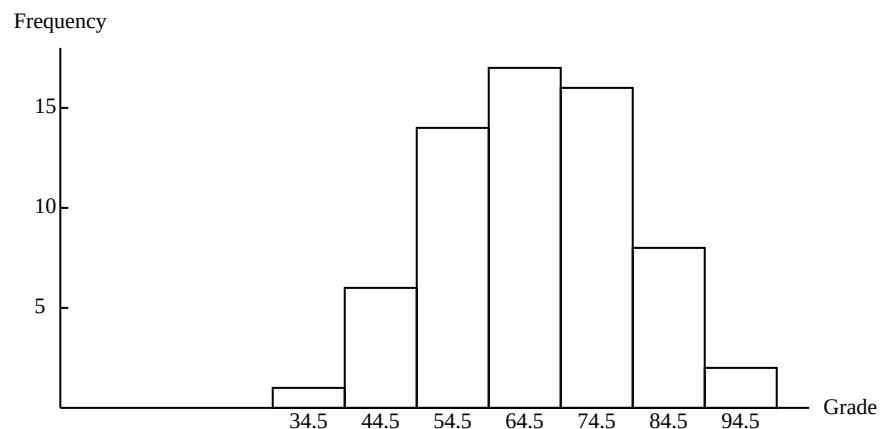
In **frequency data**, one divides the range of values into intervals and counts the number of data items that lie in each interval. For example, examination scores are given as a mark out of 100 and a schoolteacher records the number of A-plus scores (90 – 99), As (80 – 89), Bs (70 – 79), Cs (60 – 69), etc. The result is known as a **frequency distribution**. An example of a frequency distribution is given in Table 1.

Table 1. Example of frequency data.

Interval	Frequency	Cumulative frequency	Cumulative percentage
30-39	1	1	1.56
40-49	6	7	10.94
50-59	14	21	32.81
60-69	17	38	59.38
70-79	16	54	84.38
80-89	8	62	96.88
90-99	2	64	100

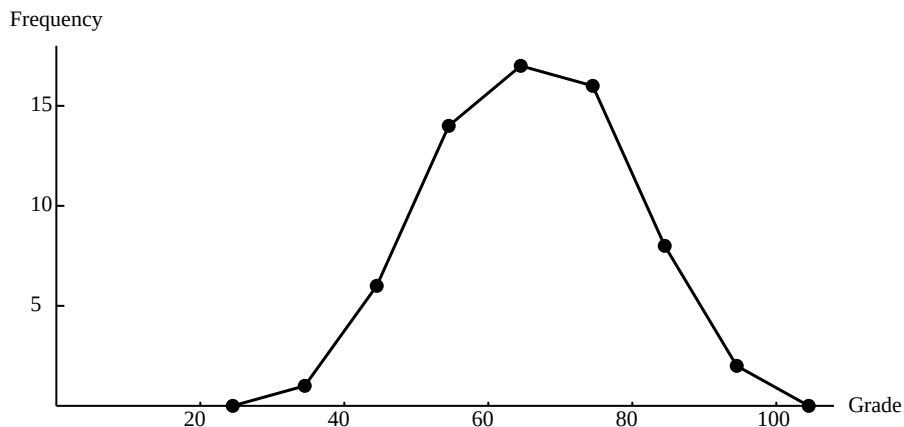
A **histogram** is used to depict a frequency distribution in which the intervals have equal size. In a histogram, one draws for each interval a rectangular bar with the base of the bar the range of that interval and the height equal to the numbers recorded. Figure 2a shows a histogram for the discrete data given in Table 1.

Figure 2a. An example of a histogram.



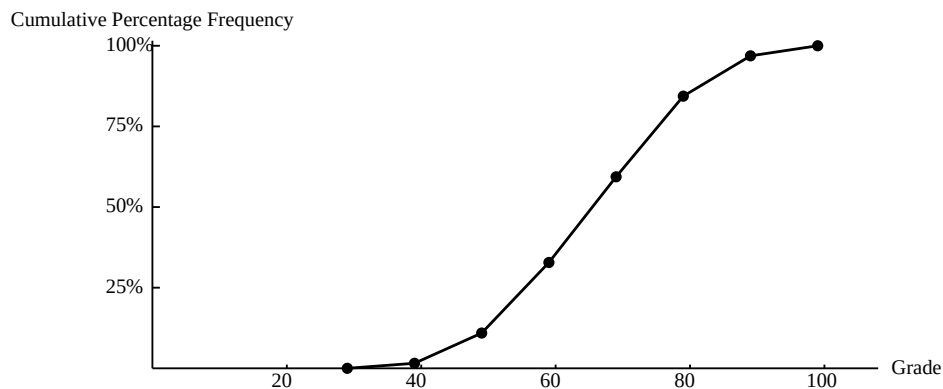
One can also depict the data with a **frequency polygon**. In this, one plots for each interval a point above the midpoint of each interval giving the number of data items recorded, and then one joins up consecutive plots (see Figure 2b). Note that the frequency polygon starts and ends at zero: one assumes that the intervals immediately before and after those given both have 0 data items.

Figure 2b. A frequency polygon.



In an **ogive** one looks at the data cumulatively. One calculates for each interval the cumulative count (or percentage)—the number of data items up to and including that interval. Then one plots for each interval a point at the maximum of the interval giving the cumulative count (or percentage). These points are then joined up to form the ogive. (If the data is discrete, then the end of an interval is halfway between the maximum of that interval and the minimum of the next interval. So in our example the ends are 39.5, 49.5, etc.) The ogive increases from 0 to total count (or 100 percent) and is often S-shaped. An example is presented in Figure 2c.

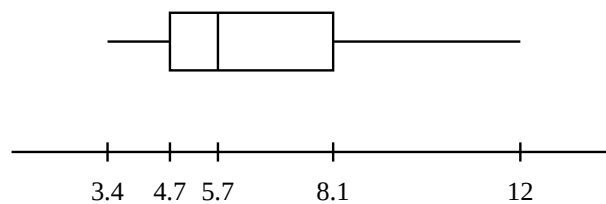
Figure 2c. An ogive.



An ogive is also useful in calculating percentiles. For example, if one wishes to know the 70th percentile, one draws a horizontal line from 70% on the Y-axis and determines where it intersects the ogive. The corresponding value on the X-axis is the desired percentile.

Another simple device is a **box-and-whiskers** diagram. This gives five statistics of the data: the minimum, the maximum, the median, the 25th percentile and the 75th percentile. An example is depicted in Figure 3. For this example, the minimum is 3.4, the 25th percentile is 4.7, the median is 5.7, the 75th percentile is 8.1 and the maximum is 12.

Figure 3. A box-and-whiskers diagram.



8.2.3 Depicting paired data

In **paired data** there is a series of observations, each with two values: say $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. For example, a chemical engineer measures the arsenic levels in a stream at various distances from a factory. He obtains a set of paired (distance, level) data. This data can be represented by a **scattergram** on X-Y axes in which there is one plot for each data point (see Figure 4). Such paired data often results from experimental research—e.g.

a physicist suspends a spring from the roof, attaches different weights to it and records the stretch for each weight.

Figure 4. A scattergram.



Based on the data, one might propose a curve which ‘fits’ the data. The data depicted in the example scattergram suggests a straight line fit. We discuss this type of problem next.

8.3 REGRESSION

Suppose a meteorologist accumulates data on temperatures at different altitudes. The question he would like answered is: is there a mathematical relationship between the two quantities (altitude and temperature); if so, what is this relationship; how strong is this relationship; and can one use the calculated relationship from the data for other temperatures?

Typically one starts with a scattergram. Then one looks at it to see whether there seems to be any pattern. This simplest case is whether there is a straight-line relationship and what the straight-line involved is. The process one uses to find the best estimate of a straight-line fit for data is known as **linear regression**. We deal with the question of how good the fit is in Section 11.1 (linear correlation).

In regression, one has a hypothesised function $y = f(x)$ which is supposed to give the relationship between x and y . The function $f(x)$ starts with some undetermined parameters: in the case of looking for the best line fit, the function is $f(x) = A + Bx$

with parameters A and B . (Recall that B is the slope of the line and A the y -intercept.) The task is to find the best values of the parameters.

In linear regression the best values of the parameters are given by the following rather intimidating formulas:

$$B = \frac{n(\sum x_i y_i) - (\sum x_i)(\sum y_i)}{n(\sum x_i^2) - (\sum x_i)^2} \quad \text{and} \quad A = \frac{(\sum y_i) - B(\sum x_i)}{n}$$

The key is to calculate, for each pair (x_i, y_i) of data, the value of x_i^2 and $x_i y_i$ and then calculate the sums. Fortunately, many calculators have a statistical mode for performing such a calculation.

Example: A study looked at the arsenic levels in a stream as a function of the distance downstream from a factory.

x (distance in km)	2	4	6	8	10	12	14
y (arsenic in mg/kl)	24.5	18.7	16.3	12.0	9.3	6.2	2.5

A scattergram for this data is given in Figure 5. This suggests a negative slope straight-line fit.

Figure 5. Scattergram for arsenic example.

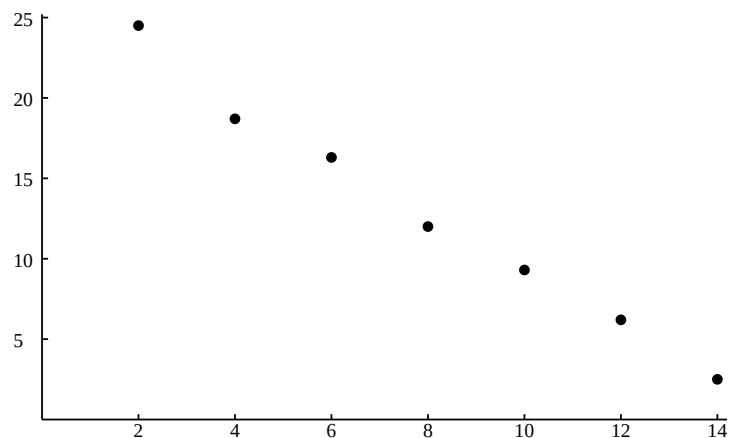


Table 2. Results for the arsenic example

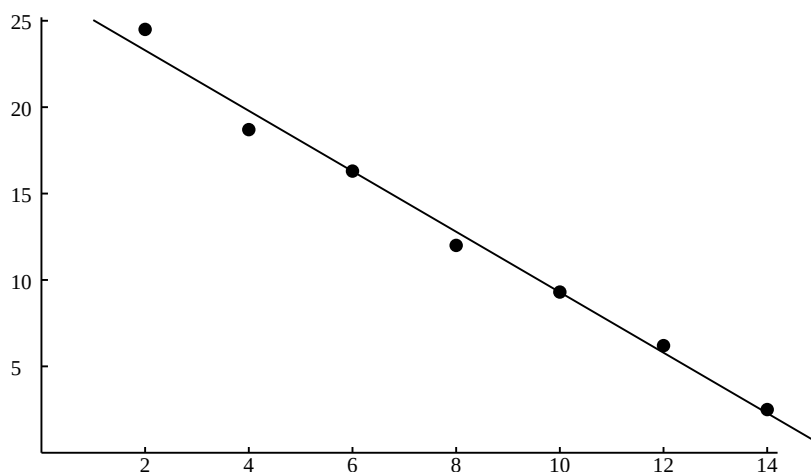
x_i	y_i	x_i^2	$x_i y_i$
2	24.5	4	49.0
4	18.7	16	74.8
6	16.3	36	97.8
8	12.0	64	96.0
10	9.3	100	93.0
12	6.2	144	74.4
14	2.5	196	35.0
56	89.5	560	520.0

Calculations give $n = 7$, $\Sigma x_i = 56$, $\Sigma y_i = 89.5$, $\Sigma x_i y_i = 520.0$, and $\Sigma x_i^2 = 560$. The formulas for linear regression give

$$B = \frac{7 \times 520 - 56 \times 89.5}{7 \times 560 - 56^2} = -1.75 \text{ and } A = \frac{89.5 - (-1.75) \times 56}{7} = 26.79$$

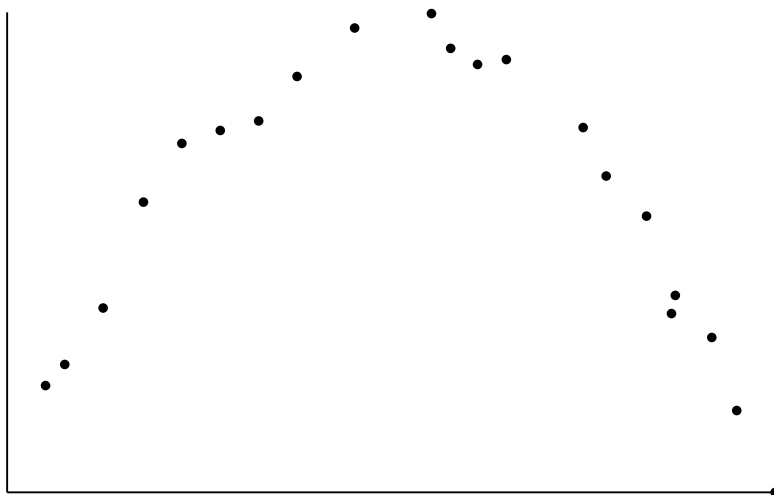
This means that the best straight-line fit is $y = 26.79 - 1.75x$. Figure 6 shows this fit.

Figure 6. Line fit on scattergram.



While linear relationships are the most common, there are many other possible relationships. For example, the data in the scattergram in Figure 7 suggests a parabolic fit. To find the best fit, one can use regression but the actual formulas are beyond the scope of the book.

Figure 7. Another scattergram.



The process of regression is guaranteed to give a fit. For example, one can fit a line to the data in Figure 7, but this is clearly inappropriate. The best fit is not necessarily a good fit. Testing whether a fit does in fact reveal a pattern in the data is discussed in Section 11.1.

Discussion questions and exercises

1. How does one summarize data? Why does one do so?
2. For each of the measures defined in the text, find a situation *(a)* where the measure is a good summary of the data and *(b)* one where it is bad.
3. Find some examples of data depiction in newspapers and scientific magazines. Are the pictures informative? Are they misleading?
4. A geographer measures the relationship between the population size of a city and the number of cinemas. Data is

population (ten thousands)	1	5	22	23	40	67
cinemas	1	1	10	10	17	28

 - (a)* Plot the data on a scattergram.
 - (b)* Find the best straight-line fit.
 - (c)* Use the line to estimate the number of cinemas for a city of 500 000 inhabitants.
5. A bacteriologist counts the number of yeast cells in a test-tube at various time intervals.

yeast	25	60	140	320	700	1605	3781	7017
time	1	2	3	4	5	6	7	8

- (a) Plot the data on a scattergram.
- (b) Describe in words a curve that would fit the data.
- (c) (For the mathematically advanced.) Find a fit of the form $y = Ae^{Bx}$. (Use the fact that if $y = Ae^{Bx}$ then $\ln y = \ln A + Bx$ and then use linear regression.)

9 Normal Distribution and Estimation

The concept of probability is the foundation of statistical tests and processes. We present a brief introduction to probability here. We then introduce the normal distribution, and explain the process of estimation.

9.1 PROBABILITY

The **probability** of an event is the proportion of time the event can be expected to occur over the long run. For a simple example, consider tossing a coin. If one repeats this 1 000 times, and heads comes up 300 times, then it appears that the probability of heads is 3/10. (Could a coin act this way? Think about it.)

The probability of an event can also be thought of as the likelihood or chance of it occurring. One classic example is rolling a die. If the die is fair then there is an equal chance that any of the six sides comes up. So the probability of a fair die coming up with number 5 is 1 out of 6, or, in symbols:

$$\Pr(D = 5) = \frac{1}{6}$$

where D stands for the value on the die.

We need some definitions: The rolling of the die is an **experiment**. (In statistics, an experiment is any action where the answer is not predetermined.) The experiment has a number of **outcomes**, and associated with each outcome is a probability. For a repeatable experiment, the probability of a specific outcome amounts to the proportion of time the outcome is likely to occur. Proportions always lie between 0 and 1, so that if something occurs 40% of the time, then its probability is 0.4. A probability of 0 means that the event never occurs, a probability of 1 means that it always occurs.

A numerical value resulting from the experiment is known as a **random variable**. A discrete random variable takes on only discrete values and often results from counting something; a continuous random variable may take on a continuous range of values and often results from measuring something. The quantity D above is a discrete random variable. For example, a geologist measures the masses of gold nuggets. This is a continuous random variable: a random variable as it is the unknown value resulting from an experiment, and continuous as it can have any real number as its outcome.

If one rolls a die 600 times, it is very very unlikely that one will get each number exactly 100 times. However, if some numbers seem to come up a *lot* more than others,

then one would suspect that the die was crooked (not fair). A simulation the authors ran produced the following frequencies of the numbers from 1 to 6: 86, 112, 122, 95, 98, 87. What do you think about the die?

The question of when to conclude that a die is crooked is precisely the type of question that statistics is designed for. And the short answer is: *if the probability of a result as extreme is sufficiently small, then one may conclude that the die is crooked.*

9.2 PROBABILITY DISTRIBUTIONS

Often one has an experiment with just two outcomes, such as tossing a coin. These outcomes are often referred to as **success** and **failure**, and the experiment as a **trial**. To obtain useful data one repeats the experiment several times and counts the number of successes.

Suppose we have an experiment with a 60% chance of success. If we perform it 9 times and get 5 successes, how likely is that? The following formula gives the answer to such questions:

$$\Pr(S = k) = \frac{n!}{k! (n - k)!} p^k (1 - p)^{n-k}$$

where:

$g! = g \times (g - 1) \times (g - 2) \times \dots \times 2 \times 1$, for g nonzero, and 1 for g zero;

S is the random variable counting the successes;

n is the number of trials;

p is the probability of success in a single trial; and

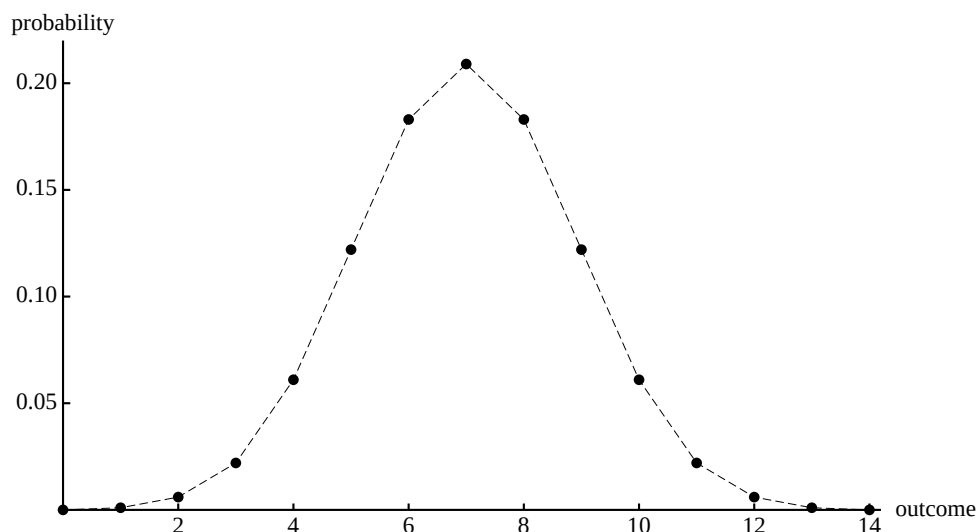
k is the desired number of successes.

In the example we need the case $n = 9$, $p = 0.6$ and $k = 5$.

The calculation says $126 \times 0.6^5 \times 0.4^4 = 25.1\%$.

The random variable S is known as a **binomial random variable**. The **probability distribution** of a random variable depicts the probability of each outcome for the variable. Figure 8 shows the probability distribution for the binomial random variable with $n = 14$ and $p = 1/2$.

Figure 8. A binomial distribution.



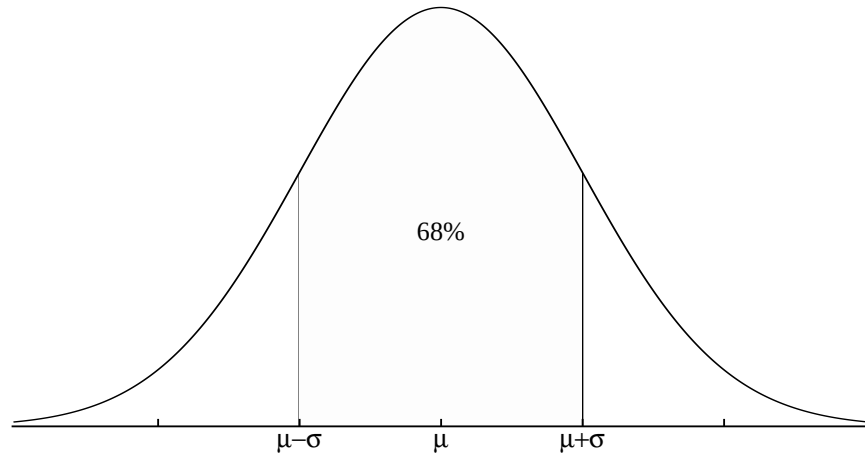
We have made one assumption which one must be explicit about: *the success or failure of one trial does not affect the (probability of) success or failure of another trial*. Statisticians would say the trials must be **independent** events. If the trials are **dependent**, i.e. not independent, then the above formula is not valid. If I toss one coin and then another coin, the results are independent. If I measure the height and mass of one child, the results are dependent.

There are other useful probability distributions, including the normal, exponential, geometric, and Poisson distributions. Of these, the normal distribution is the most important.

9.3 THE NORMAL DISTRIBUTION

The most common probability distribution for a continuous random variable is the **normal distribution**. Figure 9 shows the smooth bell-shaped curve of the normal distribution.

Figure 9. The normal distribution.



A normal distribution has two parameters: its mean and standard deviation. The distribution with $\mu = 0$ and $\sigma = 1$ is called the standard normal distribution, and the associated random variable is often denoted by Z . Many natural occurring things have a normal distribution, including heights of people, speeds of ostriches, and birth-mass of babies.

9.3.1 Using normal tables

A table of the standard normal distribution is given in Appendix B. This gives the proportion of the population that will be less than or equal to z for various values of z .

For example, suppose that the heights of South African men had a mean of 1.77 m and a standard deviation of 4 cm, and one wanted to know what proportion is less than or equal to 1.87 m tall. The method comprises two steps:

- 1) Calculate the z -value by the formula:

$$z = \frac{x - \mu}{\sigma}$$

- 2) Look up the proportion in the table in Appendix B.

For this example, $z = \frac{187-177}{4} = 2.5$. Now Appendix B is consulted. The proportion is 0.994.

If you need to find what proportion is greater than z , simply use 1 minus the proportion less than z . For example, the proportion of men that are more than 1.87m tall is $1 - 0.994 = 0.006$ or 0.6%.

9.4 STATISTICAL ESTIMATION

A fundamental problem of statistics is the estimation of measures of populations such as the population mean and standard deviation. A **point estimate** is a single-value estimate of the population measure. A **confidence interval** gives a range of values in which the measure probably lies—typically the confidence interval is such that there is either a 95% or 99% chance that the actual measure lies in the interval.

The value α (the Greek letter alpha) denotes the probability of error (that is, the measure lying outside the interval). So the probability of being correct is $1 - \alpha$. For example, for a 95% interval α is 0.05 (5%) and for a 99% interval $\alpha = 0.01$ (1%). Confidence intervals are (usually) constructed such that the probability of error is shared equally between the two sides: the probability that the actual measure is above the interval is $\frac{\alpha}{2}$ and the probability that it is below the interval is $\frac{\alpha}{2}$.

In this book we discuss the problem of estimating the population mean. The **sample mean** is denoted by $\bar{x}$ and is defined as the average of the n sample values:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The sample mean is a point estimate of the population mean μ .

To determine a confidence interval for the population mean we need, in addition to the sample mean, the **sample standard deviation**. The sample standard deviation s is defined by:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

This is similar to the formula for the population standard deviation σ except that there is the value $n - 1$ in the denominator rather than n . (There is a deep statistical reason for this difference which we do not explore here.) There is an alternative formula which is easier for calculations:

$$s = \sqrt{\frac{\sum_{i=1}^n x_i^2 - n \times \bar{x}^2}{n - 1}}$$

Example: For the sample data 8, 12, 10, 16, 7, 13,
the sample mean is 11;
and the sample standard deviation is:

$$\sqrt{\frac{782 - 6 \times 112}{5}} = \sqrt{\frac{56}{5}} = 3.35$$

In this section we discuss two common situations where one can provide confidence intervals for the mean.

9.4.1 Large sample method for estimating the mean

The large sample method can be reliably applied if there are at least 30 data values. Given some choice of α , a $1 - \alpha$ confidence interval for μ is given by:

$$\bar{x} - z \times \frac{s}{\sqrt{n}} < \mu < \bar{x} + z \times \frac{s}{\sqrt{n}}$$

In this expression, the value z is obtained from the table in Appendix C by looking up the entry in the *two-tailed* column corresponding to the chosen value of the error α . (The two-tailed values are always used for confidence intervals.)

For example, suppose one wanted to be 95% sure that the value of μ lay within the interval. Then choose α to be 5% (i.e. 0.05), and the value 1.96 is obtained for z in the column corresponding to 95% and two tailed. (Recall that one divides the chance of error equally on both sides or tails, so that there is a 2.5% chance of the actual value of μ being smaller than the calculated lower bound, and a 2.5% chance of the actual value of μ being larger than the calculated upper bound.)

Example: A biologist measures the salinity of the leaves of 60 white mangrove trees in an estuary. He wants to estimate the average salinity with a 99% confidence interval.

Procedure: He calculates the sample mean and sample standard deviation. Say he obtains $\bar{x} = 5.39$ and $s = 2.04$. He looks up the critical value for 99%, two-tailed in Appendix C. The value found is 2.58. So a 99% confidence interval for the salinity is

$$\left[5.39 - 2.58 \times \frac{2.04}{\sqrt{60}}, 5.39 + 2.58 \times \frac{2.04}{\sqrt{60}} \right] = [4.71, 6.07]$$

9.4.2 The small sample method

One can estimate the population mean from a small sample if one knows that the underlying population has a normal distribution, or one has good reason to believe

this. In this case the sample mean is related to Student's t distribution (so-called because the discoverer Gosset published the work under the pseudonym 'Student'). The process of estimation is the same as the large sample method, except that one looks up values in the t -table given in Appendix D rather than the z -table.

The t distribution has a bell-shaped curve. It has one parameter df which is known as its **degrees of freedom**. For df small the curve is a bit more peaked than the normal distribution, but for df large the two curves are indistinguishable.

A $1 - \alpha$ confidence interval (the value α denotes the probability of error) is given by:

$$\bar{x} - t \times \frac{s}{\sqrt{n}} < \mu < \bar{x} + t \times \frac{s}{\sqrt{n}}$$

where the t -value is that for $df=n-1$ degrees of freedom. The value t is found in the t -table on the row for $n-1$ degrees of freedom in the appropriate column for the α -value required (again, the two-tailed values are used).

Example revisited: In the previous example we used the large sample method to determine a 99% confidence interval for mangrove salinity. Say our biologist could only obtain 20 leaves, and again obtained a sample mean of 5.39, but a sample standard deviation of 1.84. Since he believes that the salinity has a normal distribution, he uses the small sample method. Degrees of freedom $df = 20 - 1 = 19$. $\alpha = 0.01$ (1% chance of error). Value in table is 2.861. So the confidence interval is

$$\left[5.39 - 2.861 \times \frac{1.84}{\sqrt{20}}, 5.39 + 2.861 \times \frac{1.84}{\sqrt{20}}\right] = [4.21, 6.57]$$

The above example shows that a confidence interval can be narrowed by taking n larger (think about why). This means: the larger the sample, the better the estimate.

Discussion questions and exercises

1. If you roll two honest dice, what is the probability of getting double-sixes? Any double? An odd number for the total?
2. From the population of the digits 0 through 9 take a random sample of 3 digits. Calculate μ , σ , and s .
3. Say the size of king protea flowers is normally distributed with mean 127 mm and standard deviation 13 mm. What proportion is more than 140 mm?

4. Find a 95% confidence interval for the population mean from the sample data

7.2, 9.3, 10.2, 11.4, 14.8, 16, 10.3, 11.4, 12.3, 9.9, 8

stating any assumptions you make.

5. Find a 99% confidence interval for the population mean if the sample size is 50, the sample mean is 43.41, and the sample standard deviation is 2.61.

10 Testing Hypotheses

A fundamental principle in the scientific method is the formulation of hypotheses. Once one has formulated a hypothesis, and accumulated the data, one analyses the data and then accepts or rejects the hypothesis. In this chapter we describe this process, from the formulation of the hypothesis to the test of the data.

The purpose of a hypothesis is to predict a relationship between variables that can be tested. Hypotheses direct one's research by indicating what procedures should be followed. This means that hypotheses may be reformulated *before* empirical studies start, but not *afterwards*.

10.1 HYPOTHESES, ERRORS AND TAILS

10.1.1 Hypotheses

Any statistical test revolves around the choice between two hypotheses. These are labelled H_0 and H_1 . H_0 is often called the **null hypothesis**.

We asked in a previous chapter: how would one decide if a coin was fair? In this case the null hypothesis H_0 would be that the coin is fair. The alternative H_1 is that it is biased.

10.1.2 Errors

Consider the following example: A biotechnology firm develops a new drug against tuberculosis. To test the drug, they administer the drug to some patients and a placebo to others. They then compare the results to determine whether the drug is effective or not. Suppose H_0 is that the drug has no effect, H_1 that the drug is effective.

There are two possible ways the answer could be wrong. One possibility is that one concludes that the drug is effective when in fact it has no effect. The other possibility is that one concludes that the drug has no effect when in fact it does. These are known as Type I and Type II errors respectively.

Ideally, one would like neither error to occur, but this is impossible. The smaller the risk of one type of error, the greater the risk of the other type. If one is not prepared to accept any risk of a Type I error, then one must accept a high risk of a Type II error, and vice versa. (Think about why this should be so.)

In a hypothesis test, one takes as H_0 that situation which one doesn't mind as much being accepted as true when it is in fact false, and takes an alternative as H_1 . The *benefit of the doubt goes to the hypothesis* H_0 . In other words, one focuses on the probability of a Type I error. This probability is denoted by α . The probability of a Type II error is denoted by β (beta).

Table 3: Types of errors

	H_0 true	H_1 true
H_0 accepted	okay	Type II error
H_1 accepted	Type I error	okay

10.1.3 Hypotheses and tails

An agricultural researcher wishes to test a new fertiliser, and see if it increases maize yield. There are four possible hypotheses:

H_0 : The fertiliser does not affect maize yield

H_{1a} : The fertiliser does affect maize yield

H_{1b} : The fertiliser increases maize yield

H_{1c} : The fertilizer decreases maize yield

Hypothesis H_{1a} allows for either an increase in yield or a decrease in yield—it is a combination of hypotheses H_{1b} and H_{1c} . Since there are two ways in which hypothesis H_{1a} can be true, it is termed a **two-tailed test**. Hypothesis H_{1b} and H_{1c} , on the other hand, only test for a difference in one direction (at one extreme only) and are termed **one-tailed tests**. H_{1b} is an example of an **upper-tailed** test whereas H_{1c} is an example of a **lower-tailed** test. Whenever hypothesis H_1 uses a relative term (e.g. heavier, less than) it is a one-tailed test; whenever H_1 states a difference *without* specifying the direction of the difference (e.g. unequal, affects) then the test is two-tailed.

When the test is one-tailed, the null hypothesis must be adjusted accordingly. In our example above, if H_{1b} is used then H_0 is that the fertiliser does *not increase* maize yield. Similarly if H_{1c} is used then H_0 is that the fertiliser does *not decrease* maize yield. In all cases H_0 is the *opposite* of H_1 —that is, any situation not covered by H_1 is covered by H_0 . Figures 10a, 10b and 10c show this diagrammatically.

Figure 10a. A two-tailed test.

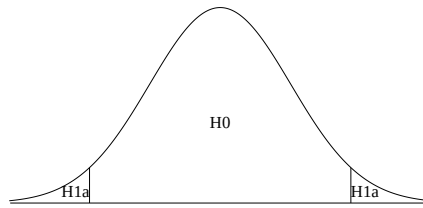


Figure 10b. An upper-tail test.

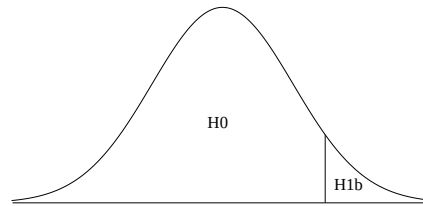
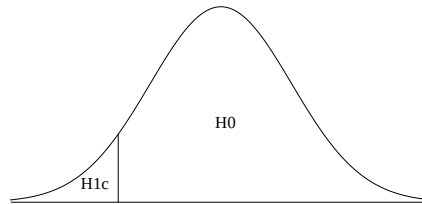


Figure 10c. A lower-tail test.



10.1.4 Level of significance

A hypothesis test is based on the probability of sample data being as extreme as the data encountered, assuming that H_0 is true. This is α , the probability of Type I error. The researcher must choose a level of significance—an upper bound for α . Common levels are 5% and 1%; for a test at the 5% level of significance there is at most a 5% chance of Type I error. In reporting the results of a hypothesis test one must state the level of significance.

10.2 THE STEPS IN A HYPOTHESIS TEST

In every statistical test, the procedure is:

1. Identify which test is appropriate.
2. Choose a level of significance α .

3. Formulate H_0 and H_1 .
4. Determine whether the test is one-tailed or two-tailed. If it is one-tailed, determine whether it is an upper- or a lower-tail test.
5. Calculate the test statistic.
6. Based on the chosen level for α , compare the test statistic with a value from a table (the **critical value**).
7. Conclude: Either accept or reject H_1 .

In particular, suppose we have a test statistic u and a table value v . Then H_1 is accepted:

- (a) in a 2-tailed test, if $u > v$ or $u < -v$.
- (b) in a 1-tailed upper-tailed test, if $u > v$.
- (c) in a 1-tailed lower-tailed test, if $u < -v$.

If H_1 is accepted, then H_0 must be rejected. However, since tests are structured so that the benefit of the doubt goes to H_0 , the converse is not true—the fact that one cannot be 95% (say) sure of H_1 does *not* mean that one is 95% sure of H_0 (think about it). So if H_1 is rejected one simply states that H_0 is *not* rejected.

10.3 TESTS OF THE MEAN OF ONE SAMPLE

One takes a sample and has a value in mind for the population mean μ . Then the question is, does the sample $\bar{x}$ contradict this value significantly? This is similar to the estimation of the mean from the sample mean described in the previous chapter.

We describe two versions:

- the large sample (z) test, and
- the small sample (t) test.

The large sample test can be used if the sample size is greater than 30, the small sample test can be used if the sample comes from a normal distribution (regardless of sample size).

In a test, the H_1 hypothesis is that the population mean is either different from (two-tailed), greater than (one-tailed upper tail) or less than (one-tailed lower tail) a particular value. H_0 is the complement of this. That is,

$$\begin{aligned}
H_0: \mu &= c, H_1: \mu \neq c; \\
H_0: \mu &\leq c, H_1: \mu > c; \text{ or} \\
H_0: \mu &\geq c, H_1: \mu < c.
\end{aligned}$$

The sample statistics are denoted by n , $\bar{x}$ and s . The large sample test uses the following test statistic:

$$z = \frac{\bar{x} - \mu}{s} \sqrt{n}$$

The test statistic calculated is compared to a value from the z -table. The small sample test uses the following test statistic:

$$t = \frac{\bar{x} - \mu}{s} \sqrt{n}$$

The test statistic calculated is compared to a value from the t -table with $df = n - 1$.

Example 1: A climatologist measures the freezing points of water taken from 42 different lakes. She wants to test whether the average freezing point is 0 degrees Celsius, and be 95% sure of her conclusion. The sample statistics are $\bar{x} = -0.49$ and $s = 1.23$.

Procedure:

1. The test is the large sample test.
2. The significance level is 0.05.
3. H_0 : the population mean is 0.
 H_1 : the population mean is not 0.
4. This is a two-tailed test.
5. The test statistic is

$$z = \frac{-0.49 - 0}{1.23} \sqrt{42} = -2.58$$

6. The table value for 5% level of significance is 1.96.
7. Accept H_1 , as $-2.58 < -1.96$. There is sufficient evidence to conclude that the population mean is not 0.

Example 2: It is known that the I.Q. of humans has a normal distribution with mean 100. It is believed that rugby players' I.Q.s are higher. Twenty members of rugby teams are selected using cluster sampling (each team

forming a cluster) and tested for I.Q. The sample mean is 102.7 and the sample standard deviation 14.8. Are rugby players' I.Q.s higher?

Procedure: The test is the small sample t -test. H_0 is that rugby players have normal or below normal I.Q. H_1 is that rugby players have higher than normal I.Q. This is a one-tailed upper tail test. The calculated test statistic is $t = 0.816$. The degrees of freedom $df = 19$. Value in the table is 1.73 for 5% level of significance. As $0.816 < 1.73$, at 5% significance level one cannot conclude that rugby players have higher I.Q.s.

10.3.1 Estimation and hypothesis testing

These hypothesis tests of the population mean are linked to the estimation of population mean discussed in Section 9.4. For example, in a two-sided test, H_0 is rejected if and only if is outside the confidence interval.

10.4 A TEST FOR EQUIVALENCE OF RELATED SAMPLES

Related sample pairs can be formed in two ways. One is where two sets of measurements are compared (for example, the results of 'before and after' tests on the weights of mice). The other is where the data is gathered from measurements of matched samples (see Section 5.4).

To perform the test on pairs (a_i, b_i) , one calculates the difference data $d_i = a_i - b_i$. This difference data is itself a sample. For a two-tailed test, the null hypothesis H_0 is that the two population means are equal. This is the same as testing whether the population mean μ_D of the difference data is zero, i.e. H_0 is that $\mu_D = 0$. Hypothesis H_1 is that μ_D is non-zero. We use the difference data for a t -test. The sample statistics for the difference data are denoted n , $\bar{x}_d$ and s_d . Note that for matched data n denotes the number of data *pairs*.

The t -test for equivalence of related samples uses the following test statistic:

$$t = \frac{\bar{x}_d}{s_d} \sqrt{n}$$

Degrees of freedom: $df = n - 1$.

Example: An educator proposes that using multimedia would improve the teaching of calculus to first-year students. So she picks a sample group of 20

students at random. Students are matched in terms of matric mathematics symbols into 10 pairs, with one member of each pair being placed in the experimental group and the other in the control group. The measurements she uses as data are the final calculus exam marks of the students.

Procedure: The test is the t -test for equivalence of related samples. It is decided to do the test at both the 1% and the 5% levels of significance. Hypothesis H_0 is that the multimedia has no effect on or worsens the calculus results. Hypothesis H_1 is that the multimedia improves the calculus results. The test is one-tailed upper tail. The measured data is tabulated.

Table 4: Results for the calculus example

Pair	Marks Group Experiment	Marks Group Control	Difference	d_i^2
a	18	16	2	4
b	16	15	1	1
c	14	15	-1	1
d	16	16	0	0
e	17	10	7	49
f	14	10	4	16
g	12	15	-3	9
h	13	8	5	25
i	11	11	0	0
j	9	5	4	16
Totals			19	121

Calculations: $\bar{x}_D = 1.9$ and $s_D = 3.07$ so that test statistic is $t = 1.9 \frac{\sqrt{10}}{3.07} = 1.96$. The number of degrees of freedom is $df=9$, so the critical values are 1.83 and 2.82. So one can conclude an improvement at the 5% level of significance but not at the 1% level of significance. (The educator can be 95% certain that H_1 is true, but not 99% certain.)

Notes:

- The necessary assumptions for hypothesis tests must be checked. If not known beforehand to be true, this may involve further statistics tests!
- It is sometimes advisable to estimate β , the chance of a Type II error.

Discussion questions and exercises

1. Explain the meaning of: hypothesis, null hypothesis, tail, level of significance, hypothesis test, test statistic, critical value.
2. In drug testing the null hypothesis is normally taken that the drug has no effect and the alternative is that the drug is effective.
 - (a) What are Type I and Type II errors?
 - (b) Which is preferable?
 - (c) What about a disease such as HIV/Aids which is rampant and for which no cure has been found yet? Should one deny people the drug just because one is not totally convinced that it is effective? Discuss.
3. The sample data of Question 4 from Chapter 9 is believed to come from a population with mean 12.8. Test the hypothesis that:
 - (a) this is not true; and
 - (b) the actual mean is smaller than 12.8.
4. If we always use a 0.05 level of significance, does this mean that on average 1 out of 20 conclusions will be wrong?
5. A quantity surveyor calculates the amount of material needed for a project as 60 units. For a sample of 40 projects she obtains a mean of 65.32 units and a standard deviation of 3.17 units. Test to see whether the sample data conforms with her calculations.
6. A student makes a series of measurements of the same compound on two scales in milligrams and obtains the following results:

Scale 1:	473	321	555	678	299	407
Scale 2:	472	320	556	682	307	415

 - (a) Find a 95% confidence interval for the average difference between the two scales.
 - (b) Test the belief that scale 2 gives higher readings than scale 1.
 - (c) What does this say, if anything, about the validity and reliability of the two scales?

11 More Hypothesis Testing

In this chapter we describe some frequently-used tests. First we describe a test for linear correlation, and then introduce the concept of partial correlation. Then we describe two more tests: the chi-square test for patterns in frequency data and the test for the equality of means for independent samples.

11.1 A TEST FOR LINEAR CORRELATION

In Section 8.3 we described how to find the best line fit for paired data. Determining whether there is in fact a linear relationship requires another hypothesis test. Pearson's product-moment **coefficient of linear correlation** is calculated by the formula:

$$r = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{\sqrt{n \sum x_i^2 - (\sum x_i)^2} \sqrt{n \sum y_i^2 - (\sum y_i)^2}}$$

(Wow!) This parameter lies between -1 and 1 . A value of 1 indicates a perfect linear dependence with positive slope. (An increase in the value of variable X is associated with a proportionate *increase* in the value of variable Y .) A value of -1 indicates a perfect linear dependence with negative slope. (An increase in the value of variable X is associated with a proportionate *decrease* in the value of variable Y .) A value of 0 or thereabouts says very little.

The hypothesis test: H_1 can be that either

- (a) there is a linear correlation (two-tailed),
- (b) there is a positive slope correlation (one-tailed upper tail), or
- (c) there is a negative slope correlation (one-tailed lower tail).

The hypothesis H_0 is, as usual, the complement of H_1 . The test statistic is the Pearson coefficient of linear correlation r . The test statistic is compared with the critical values obtained from the table in Appendix E, with $n - 2$ degrees of freedom (where n is the number of data pairs), and value desired of α .

Example: In Section 8.3 we considered a chemical engineer who obtained the following measurements for arsenic levels at different distances from a factory. The engineer wishes to test whether the arsenic distances decrease linearly the further one is from the factory.

x (distance in km)	2	4	6	8	10	12	14
y (arsenic in mg/kl)	24.5	18.7	16.3	12.0	9.3	6.2	2.5

Procedure: This is a one-tailed lower tail test with H_0 : arsenic levels do not decrease linearly with distance from factory, and H_1 : arsenic levels do decrease linearly with distance from the factory.

Calculations give $n = 7$, $\Sigma x_i = 56$, $\Sigma y_i = 89.5$, $\Sigma x_i y_i = 520.0$, $\Sigma x_i^2 = 560$ and $\Sigma y_i^2 = 1490.81$. Hence

$$r = \frac{7 \times 520 - 56 \times 89.5}{\sqrt{7 \times 560 - 56^2} \sqrt{7 \times 1490.81 - 89.5^2}} = -0.995$$

The $df = 7 - 2 = 5$. Table values are 0.669 (5% level) and 0.833 (1% level). So the engineer can accept H_1 at the 1% level (he can be 99% certain that H_1 is true).

11.2 PARTIAL CORRELATION

It is important to note that the existence of high linear correlation between two variables does not necessarily imply a cause-and-effect relationship between the two variables. It is likely that ownership of motor vehicles and ownership of television sets have a high correlation. This does not mean that owning a car *causes* someone to own a television set. Some other factor could be at work, e.g. wealth.

Partial correlation tries to remove the effect of the third variable and thereby find the ‘true’ relationship between the first and second variables. The formula for partial correlation is:

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{1 - r_{13}^2} \sqrt{1 - r_{23}^2}}$$

where:

$r_{12.3}$ is the partial correlation between variables 1 and 2 with the effect of variable 3 removed;

r_{12} is the correlation between variables 1 and 2;

r_{13} is the correlation between variables 1 and 3;

r_{23} is the correlation between variables 2 and 3;

and correlation is measured by the Pearson coefficient of linear correlation (see previous section).

To continue our example, say a researcher found that owning TVs (variable 1) had a correlation of 0.7 with owning cars (variable 2). It was also found that the correlation of TV ownership with wealth (variable 3, which the researcher defined as assets over R50 000) was 0.8, and the correlation of car ownership with wealth was 0.9. The formula gives:

$$r_{12.3} = \frac{0.7 - 0.8 \times 0.9}{\sqrt{1 - 0.8^2} \sqrt{1 - 0.9^2}} = -0.08$$

In other words, the researcher finds almost no correlation at all (-0.08 is a tiny negative correlation) between TV ownership and car ownership once the effect of wealth has been removed.

11.3 THE CHI-SQUARE TEST

In this section we describe the chi-square test for dependence of two qualitative variables. For example, the colour of a particular chemical solution can be red, orange, purple or blue and its acidity can be categorised as high, medium or low. We wish to know if colour and acidity are related.

Say the first qualitative variable divides the entire population up into R classes and the second divides the entire population up into C classes. In our example above $R = 4$ and $C = 3$. To apply the chi-square test, one takes a sample of the population, and categorises each item according to both variables. The sample items are divided into R classes according to the first variable and into C classes according to the second.

The data is represented in a **contingency table**. Each row corresponds to a different class of the first variable and each column to a different class of the second variable. The entry o_{ij} in row i and column j gives the number of sample items which are classified as class i for the first variable and class j for the second. Table 5 shows the observed frequencies for our example. Totals for each row and column are calculated, as well as the overall total.

Table 5: Observed frequencies

	High	Medium	Low	row total
Red	15	8	5	28
Orange	12	9	9	30
Purple	25	24	31	80
Blue	6	14	20	40
Column Tot	58	55	65	178

Alongside that table, one constructs the **expected contingency table** which would result if the two variables were independent: To calculate the entry e_{ij} in row i and column j of the expected contingency table, one multiplies the total in row i by the total in column j and divides the result by the overall total. The expected contingency table for our example is given in Table 6.

Table 6: Expected frequencies.

	High	Medium	Low
Red	9.12	8.65	10.22
Orange	9.78	9.27	10.96
Purple	26.07	24.72	29.21
Blue	13.03	12.36	14.61

In the test, the hypothesis H_0 is that the two variables are independent and H_1 is that they are dependent. The test statistic is:

$$\chi^2 = \sum \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

where o_{ij} denotes the observed frequency in row i , column j , and e_{ij} the expected frequency in row i , column j , and the summation is over all entries in the contingency table.

For the above data, the contribution to the test statistic χ^2 is as follows:

Table 7: The contribution to the test statistic

3.785	0.049	2.670
0.506	0.008	0.349
0.044	0.021	0.109
3.796	0.218	1.991

For example: the entry 2.670 is calculated by $\frac{(5-10.22)^2}{10.22}$.

One needs to compare this with the chi-square tables. The degrees of freedom for a χ^2 test are given by $(r-1)(c-1)$. In our example above, which is a 4-by-3 table, this is $df = (4-1)(3-1) = 6$. The concept of a lower tail does not exist for the χ^2 test statistic as the statistic is always positive. As all the error is thus in the upper tail, a two-tail test becomes an upper-tailed test.

The null hypothesis H_0 is that colour and acidity are independent, and H_1 is that colour and acidity are dependent. The test statistic $\chi^2 = 13.546$ with $df = 2 \times 3 = 6$. So critical values are 12.59 (5%) and 16.81 (1%). Accept at the 5% level of significance but not at the 1% level of significance.

There are limitations on the use of this test. An important rule of thumb is that every entry in the expected contingency table must be at least 5.

The chi-square test is an example of a **non-parametric test**. Such tests can be used when the data is qualitative or discrete as they do not assume anything about the underlying distribution of the population. Such tests are especially useful if the data is ordinal rather than cardinal (i.e. the data items are ranked from 1 to n rather than having values showing quantity).

11.3.1 Comparing one qualitative variable with a theory

A version of the chi-square test can be used where one has frequency data and wishes to check whether this conforms with a theorized distribution. For example, one might wish to test whether data comes from a random distribution or not.

There is one qualitative variable. Say this divides the population into M classes. The data gives the observed frequencies o_i and the theorized distribution gives the expected frequencies e_i . Then the test statistic is given by a similar formula to the one above:

$$\chi^2 = \sum \frac{(o_i - e_i)^2}{e_i}$$

There are $M - 1$ degrees of freedom.

Example: The crooked die story completed. Recall that we rolled a (simulated) die 600 times and got 86, 112, 122, 95, 98, 87. Is this die crooked?

Procedure: Hypothesis H_0 is that the die is fair, hypothesis H_1 that it is crooked. Expected values are 100 each. The test has 5 degrees of freedom. The test statistic $\chi^2 = 10.22$ (calculation is $\frac{(-14)^2}{100} + \frac{12^2}{100} + \frac{22^2}{100} + \frac{(-5)^2}{100} + \frac{(-2)^2}{100} + \frac{(-3)^2}{100}$). Testing at the 5% level, we find that the table value is 11.07. So we cannot reject the hypothesis that the die is fair.

11.3.2 Tests with one degree of freedom

In general, the chi-square test is a test of dependence. However, in the special case where $df = 1$, one can test for a one-tailed hypothesis. One case is a 2-by-2 contingency table. For example, one may be testing a new drug with two categories: drug and placebo, and with two results recover or not recover. One can use a chi-square test to test whether the drug is effective or not. Another case is illustrated in the next example.

Example: An economist wrote a computer program to forecast whether tomorrow's JSE industrial average would be higher than today's. She ran it for three months. It is known that an increase occurs on average 3 days out of 5. As the program's forecast was right only 66 days out of 90, a cynic suggests that one might as well toss a coin which comes up heads 60% of the time. Is he correct?

Procedure: This test is the chi-test for one qualitative variable. H_0 is that $\mu = 0.6$ and H_1 is that $\mu > 0.6$. This is a one-tailed test. The observed frequencies are 66 (correct) and 24 (incorrect). The expected frequencies are $(3/5) \times 90 = 54$ (correct) and $(2/5) \times 90 = 36$ (incorrect). The test statistic $\chi^2 = \frac{(66-54)^2}{54} + \frac{(24-36)^2}{36} = 6.67$. The value in the table is 2.71 (5% level) and 5.51 (1% level). Therefore, one can conclude with 99% confidence that the program is worthwhile.

11.4 TESTS FOR EQUALITY OF MEANS (INDEPENDENT SAMPLES)

Another situation one may encounter is where one has two independent samples from different populations but wonders whether the means are the same.

We describe two tests. These are the large sample (z) test and the small sample (t) test. The t -test can only be applied if both samples come from normal populations, and the standard deviations of the normal populations are similar. (The equality of standard deviations can be checked by using what is known as the F -test.)

In both tests there are two samples which we denote by A and B . The null hypothesis H_0 in a two-tailed test is that the two population means (which we denote by μ_A and μ_B) are equal. The sample statistics for A are denoted by n_A , $\bar{x}_A$ and s_A , and for B are denoted by n_B , $\bar{x}_B$ and s_B .

The z -test for equality of means (large samples) uses the test statistic:

$$z = \frac{\bar{x}_A - \bar{x}_B}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}}$$

The t -test for equality of means (normal populations) uses the test statistic:

$$t = \frac{\bar{x}_A - \bar{x}_B}{\sqrt{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}} \sqrt{\frac{n_A n_B (n_A + n_B - 2)}{n_A + n_B}}$$

with $df = n_A + n_B - 2$ degrees of freedom.

Large sample example: A large software house uses the languages C and Natural for programming. They wish to compare the development time as they believe that Natural is faster to program in. They know that recently they have completed 42 projects in Natural and 30 in C. The sample statistics in person-months are $\bar{x}_N = 3.3$, $s_N = 1.1$, $\bar{x}_C = 4.0$ and $s_C = 0.8$.

Procedure: The test is the two-sample z -test. H_0 is that Natural is the same or worse than C, and H_1 that Natural is better. This is a one-tailed lower tail test. Test statistic: $z = -3.13$. Critical values -1.64 (5%) and -2.33 (1%). As the test statistic is less than both critical values, one can conclude with 99% certainty that Natural is quicker to develop.

Small sample example: A zoologist has developed a new nutrient which when taken as a supplement by rats is believed to increase weight by at least 4 grams. He tested the nutrient on rats using an experimental group of 10 rats and a control group of 10 rats. He has good reason to believe weight gain to be normally distributed. In the control group the gain statistics in grams were $\bar{x}_C = 37.3$ and $s_C = 4.2$. In the experimental group they were $\bar{x}_E = 44.6$ and $s_E = 4.7$.

Procedure: The test is the two-sample t -test. H_0 is that the nutrient increases mass by 4 g or less, and H_1 that the nutrient increases mass by more than 4 g. This is a one-tailed test (upper tail). Test statistic: $t = 1.66$. Degrees of freedom: $df=18$. Critical values from the table are 1.73 (5% level) and 2.55 (1% level). Therefore one can be 95% certain that the nutrient increases weight by more than 4 grams, but not 99% certain.

There is also an extension of these tests to a null hypothesis H_0 in a two-tailed test of the form $\mu_A - \mu_B = C$. The test statistics are the same as those in the above formulas, except that the numerator $\bar{x}_A - \bar{x}_B$ is replaced by $\bar{x}_A - \bar{x}_B - C$.

Discussion questions and exercises

1. What is linear correlation? What is negative linear correlation? Is it possible to have a negative linear correlation without a linear correlation? A linear correlation without a negative one? A negative correlation without a negative linear correlation?
2. Test the data of Question 4 of Chapter 8 for linear correlation.
3. Pick 10 number pairs at random. Each number in each pair should be a whole number in the range 1 to 10. Is there a linear correlation among the number pairs chosen?
4. The drug company finally does the trial on a new treatment for tuberculosis. There are 32 patients who complete the trial. Of the 18 patients on placebo, 8 recover and 10 do not. Of the 15 patients on the drug, 12 recover and 3 do not.
 - (a) Is there any difference between the effectiveness of the drug and that of the placebo?
 - (b) What does your result in (a) tell you about the drug?
5. According to Mendel's laws of genetics, the population of plants produced from a certain crop of seedlings should be red flower green stem, orange flower yellow stem, red flower yellow stem, orange flower green stem in the proportion 9:3:3:1. A random sample of 100 seeds are sprouted and the observed frequencies are 52:19:24:5. Are these results compatible with the laws?
6. In a case-study, a researcher into poverty measures the correlation between income and IQ and obtains $r = 0.6$. She also measures the correlation between income and years of schooling, obtaining 0.8, and between years of schooling and I.Q., obtaining 0.55. Use partial correlation to compute the correlation of income and I.Q. without the influence of schooling. What does your result mean?
7. An economist measures the prices of samples of 40 widgets in both Ankara and Bisho in rands. In Ankara he obtains a mean of 14.31 and standard deviation 3.12 and in Bisho a mean of 14.64 and standard deviation 3.03. Is there a significant difference in the price?
8. Two samples from different normal populations yield the following sample statistics: $n_A = 6$, $\bar{x}_A = 100$, $s_A = 10$, $n_B = 8$, $\bar{x}_B = 80$, and $s_B = 9$. Test for inequality of population means.